

Assignment Instructions: BFV Homomorphic Encryption and Ciphertext Addition

Objective

This assignment is designed to give you hands-on experience with Homomorphic Encryption. You will learn how to use the BFV encryption scheme provided by the TenSEAL library to perform a simple ciphertext addition. You will encrypt the last five digits of your student ID using a public key and perform ciphertext addition with a randomly encrypted number provided by the TA, then upload the resulting ciphertext.

Files Provided by the TA

The following files will be provided:

Filename	Purpose	Download Link (if provided)
public.key	Public key used for encryption and ciphertext addition	Public key link
[your_student_id].tenseal	A randomly encrypted number (each student receives a different one)	Ciphertext link

! Do not modify these files or exchange them with others.

What You Need to Do

1. Write a Python script to load the provided `public.key` and `[your\_student\_id].tenseal`.
2. Convert the last five digits of your student ID to an integer and encrypt it as a single-element ciphertext using BFV.
3. Add your encrypted number with the provided ciphertext using homomorphic addition.
4. Save the result as a file named: `[your\_student\_id]\_enc\_result.tenseal`.

Example

- Your student ID is 410785021 → Last five digits: 85021
- Suppose the TA provides a random encrypted number representing [85684]
- You encrypt 85021 and add it to the random encrypted number: $[85021] + [85684] = [170705]$
- You will not see these actual values, but you will complete the ciphertext addition.
- Output file should be named: 410785021_enc_result.tenseal

Submission Requirements

Please upload the following file:

- `[your\_student\_id]\_enc\_result.tenseal`

! Incorrect filenames, wrong ciphertexts, or duplicate content may be considered invalid or plagiarism.

Installation

Make sure to install TenSEAL in your environment using the following command:

```
pip install tenseal
```

If you encounter any installation or execution issues, please contact the TA as soon as possible.

Deadline

Submit your ciphertext file to NTU COOL before 11:59 PM on Friday, November 14, 2025.